Detector-coupled Monte Carlo estimate of background and unresolved-line sensitivity for a balloon-borne 511 keV Laue-lens TES telescope

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Abstract

The Galactic 511 keV annihilation signal is dominated by diffuse bulge and disk emission, while compact or point-like contributions remain poorly constrained. We present a detector-coupled Monte Carlo estimate for a balloon-borne pointed telescope combining a 10 m Ge(111) Laue lens with a transition-edge sensor microcalorimeter focal plane. The signal chain transports focused 511 keV photons through the Be aperture into the detector/cryostat mass model. The background chain transports prompt atmospheric particles and photons and production-position-sampled delayed activation decays through the same model. Prompt, delayed, and signal streams are analyzed with a common 510.79–511.21 keV window, CsI anticoincidence veto, and topology/field-of-view consistency selection. For a reference point-source flux of $10^{-4}\,\mathrm{ph\,cm^{-2}\,s^{-1}}$, the selected day-15 signal and background rates are $1.19 \times 10^{-3}\,\mathrm{s^{-1}}$ and $3.92 \times 10^{-2}\,\mathrm{s^{-1}}$, respectively; prompt atmospheric events account for 93.4% of the selected background. Folding these rates through the adopted synthetic 20 d trajectory gives $Z_{20\mathrm{d}} = 7.80$ and a corresponding 3σ flux threshold of $3.85 \times 10^{-5}\,\mathrm{ph\,cm^{-2}\,s^{-1}}$. The result is a selected-rate flight-sensitivity estimate for a spectrally unresolved compact 511 keV source under the reference geometry and response model. Prompt atmospheric events dominate the selected background, while the delayed component is smaller but materially structured, identifying shielding, side-entry collimation, local materials, and activation validation as the main next engineering and analysis targets.

Keywords Gamma-ray instrumentation, 511 keV annihilation line, transition-edge sensors, Laue lens, balloon-borne telescope, activation background

In this work we develop an end-to-end detector-coupled Monte Carlo chain for this concept and use it to estimate the in-flight background and point-source performance at balloon altitude and to provide directional guidance for design optimization. The chain takes the EXPACS/PARMA atmospheric cosmic-ray field as input and, at balloon altitude, transports the prompt particles while tracking their activation products, yielding both the prompt atmospheric background and the delayed activation background that builds up over the flight; the focused 511 keV reference signal is provided by a 10 m Ge(111) Laue-optics model. The signal and both backgrounds are injected into the same representative balloon-borne TES focal-plane mass model (a dilution-refrigerator cold end, passive shielding, and an active anticoincidence shield), processed with a common narrow energy window, anticoincidence veto, and topology/field-of-view selection, and folded along a reference flight trajectory to give the selected signal rate, the background composition, and the counting significance. The selected background is dominated by the prompt atmospheric component, so the shield geometry, side-entry collimation, and shield material are the most direct directions for design optimization; the delayed activation and the upstream optics-hardware background enter the same evaluation as secondary components. This work thereby estimates the in-flight background and detection performance of the balloon platform and indicates directions for design optimization.

0.1 Science and instrument requirements

The simulated instrument is aimed at 511 keV point-source or unresolved-central-excess components from the Galactic-center region that have not been detected by previous instruments. Existing INTEGRAL/SPI measurements have established a 511 keV sky dominated by diffuse bulge and disk emission and limit point-like 511 keV sources superposed on that diffuse emission to fluxes of order $10^{-4}\,\mathrm{ph\,cm^{-2}\,s^{-1}}$ [Siegert et al.(2016), Yoneda et al.(2025)]; an independent INTEGRAL/IBIS compact-source search also found no Galactic 511 keV point sources and gives an upper limit of about $1.6 \times 10^{-4}\,\mathrm{ph\,cm^{-2}\,s^{-1}}$ for a $\sim$ 10 Ms Galactic-center exposure [De Cesare(2011)]. For a representative 511 keV atmospheric transmission of $T \simeq 0.739$ at balloon altitude, this limit corresponds to an above-payload photon-flux scale of about $1.2 \times 10^{-4}\,\mathrm{ph\,cm^{-2}\,s^{-1}}$. The simulation therefore adopts $F_0 = 10^{-4}\,\mathrm{ph\,cm^{-2}\,s^{-1}}$ as the reference point-source photon-flux scale. Compared with the wide-field spectroscopic and coded-mask searches performed by INTEGRAL, the modeled payload adds Laue focusing to improve the point-source response and uses a high-energy-resolution, multilayer thick-absorber TES microcalorimeter array as the focal-plane detector to sharpen the 511 keV line measurement [Shirazi et al.(2023), Caputo et al.(2017)].

This work first considers a balloon payload as an early test platform, with the expectation that the approach can later be extended to a satellite observing platform.

This goal imposes four requirements on the instrument and the simulation: first, the telescope must concentrate 511 keV photons onto a small focal plane, so that a larger solid angle can be covered without sacrificing collecting area. Second, the focal-plane detector must provide sub-keV energy response sufficient to resolve detailed line structure. Third, the detector and shield model must suppress atmospheric, activation, and side-entry backgrounds. Fourth, the simulation chain must treat prompt background, delayed activation, focused signal, vetoes, event topology, and mission-time variation consistently, so that signal and background are evaluated under the same detector selection.

The remainder of this paper is organized as follows. Section 1 describes the detector and cryostat mass model; Section 2 the simulation framework—the workflow, optics coupling, background sources, event selection, and mission-time fold; Section 3 the selected-rate and sensitivity results together with the normalization, convergence, and boundary checks; and Section 4 the discussion and conclusions.

1 Detector and cryostat mass model

The detector mass model is an engineering Geant4/MEGAlib proxy mass model that contains a balloon-borne multilayer TES array mounted at the cold end of a compact dilution refrigerator (DR) [Shirazi et al.(2023), Gottardi & Nagayoshi(2021)]. The detector model also contains typical DR cryostat structures, active shielding, passive shielding, magnetic shielding, side-entry windows, collimation structures, copper thermal structures, and other cryogenic service-mass proxies. Figure 1 shows two-dimensional views of the mass model used in this paper.

The TES detector concept uses an absorber-coupled AlMn film structure developed in our laboratory; the advantages of AlMn alloy include transition-temperature tunability through composition and annealing, suitability for array fabrication, and relatively low magnetic-field sensitivity [Xie et al.(2026), Vavagiakis et al.(2017)]. In the present concept, the TES thermometer film is an AlMn film of order 200 nm thickness, whereas the γ-ray absorber is a millimetre-scale metal body. The absorber uses $1.5 \times 1.5 \times 3.0\,\mathrm{mm}^3$ Ta pixels; the high atomic number of Ta increases the interaction probability for 511 keV photons, and a 3.0 mm thickness corresponds to about a 49% single-layer detection efficiency at 511 keV, so the six-layer array can provide near-unity detection efficiency. In the current geometry, the center-to-center spacing between adjacent Ta array layers is 12 mm; this spacing is designed with the efficiency of the subsequent Compton-based kinematic-track veto mechanism in mind. The choice of Ta also considers its low-temperature low-specific-heat properties as a superconductor (this paper uses an order-of-magnitude estimate of $10^{-2}\,\mathrm{J\,K^{-1}\,m^{-3}}$) and a relatively high photoelectric-to-Compton interaction ratio; based on NIST XCOM cross sections near 511 keV, this ratio is about 0.731 [Hubbell & Seltzer]. Based on laboratory experience and by substituting the current Ta absorber heat capacity of $67.5\,\mathrm{pJ/K}$, the simulation adopts $\Delta E_{\mathrm{FWHM}} \simeq 420\,\mathrm{eV}$ at 511 keV as the current simulated energy resolution.

Because the TES film and Ta absorber differ greatly in size and mass, and because the TES film itself does not contribute significant 511 keV interactions, the transport geometry uses the Ta absorber to represent the main sensitive volume of a TES pixel. The explicit sensitive volume has six layers, 376 Ta absorber pixels per layer, and 2256 pixels in total, with 1.5 mm length and width and 3 mm thickness. A Si substrate proxy is placed below each Ta pixel, using Si to represent the SiO2-SiNx-Si substrate because this simulation does not model detailed thermal conduction. TES-film, wiring, and readout details are not built layer by layer as explicit interaction volumes; instead, they enter the model through the assumed energy response and service-mass proxies. In the current manuscript, the TES energy resolution is used as the 511 keV line-window width, with W511 = 511 keV $\pm$ 210 eV, i.e. 510.79–511.21 keV.

The TES array is mounted on the required copper support and thermal-link structures. Copper provides the decisive thermal-conduction path; a copper thermal finger exits through the bottom hole of the sample box

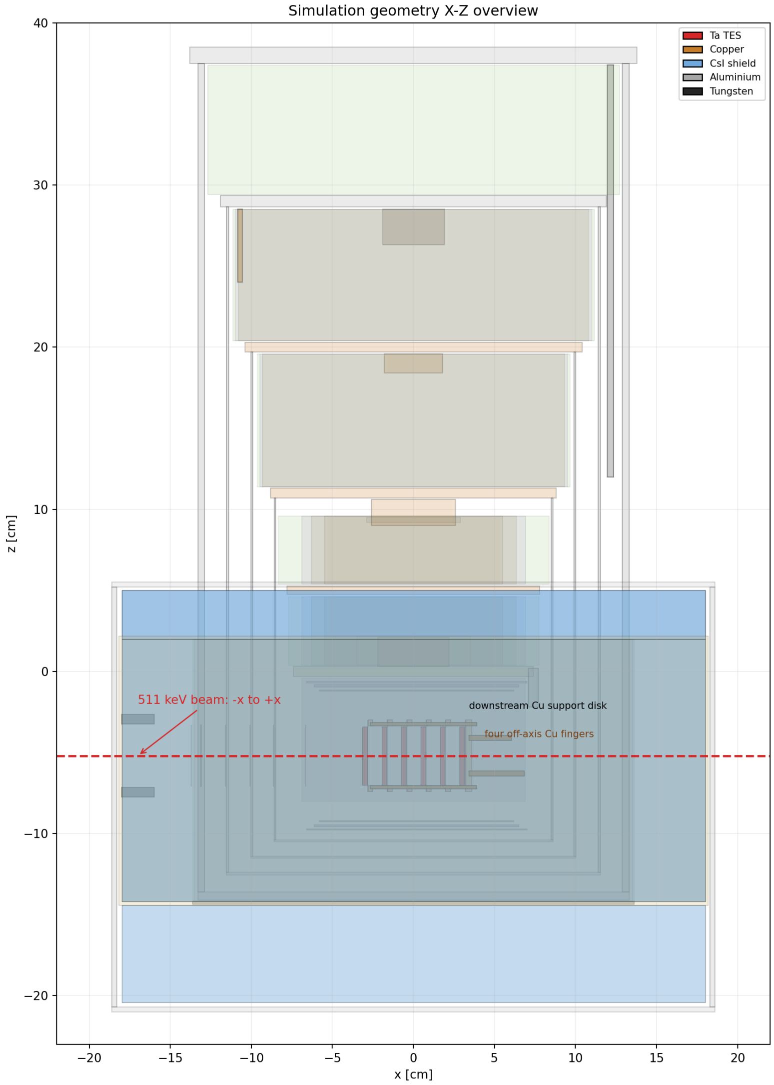

Figure 1: Two-dimensional overview of the detector/cryostat proxy mass model, showing the X–Z view of the DR tower, detector bay, active shield, and 511 keV side-entry beam path.

and connects to the 50 mK cold plate. The near-field TES sample box serves as both magnetic shielding and local passive shielding. The current mainline geometry consists of a 2 mm Nb inner layer and a 2 mm MuMetal outer layer, with MuMetal represented in the material table as Ni:Fe = 4:1. The sample-box entrance window is covered by a thin Al window; the rear end cap retains the hole for the cold finger. A W–Nb double-layer sample box is treated as a comparison passive-shielding + magnetic-shielding material design; the X-ray window is opened on the side of the cryostat, with a 300 μm thick Al window embedded in the thermal-shield shell inside each cryostat layer, a 25 μm thick Be window placed in the vacuum shell, and an x-thick W collimator placed outside the Be window, with the collimator opening strictly aligned to the TES pixels. The side window is strictly defined through Boolean operations. The mass model also introduces scintillator active shielding, wrapped around the outer side wall, bottom, and top of the cryostat. Segmented CsI active shielding is used, with a radial side-shield thickness of 4 cm; the bottom and top shields are included in the geometry as axial end volumes. CsI volumes enter the simulation as transport and activation materials, and their deposited energies are read in post-processing to apply the active veto; the counting threshold is 50 keV. BGO active shielding is treated as a material-comparison case; the BGO threshold is 70 keV. Considering the physical layout of the DR and the side-entry optical path, the cryostat instrument frame is rotated as a whole by 45 degrees, so that the local side window is tilted $45°$ upward in the zenith-frame source coordinates.

2 Simulation framework

Detector-coupled Monte Carlo transport through detailed mass models is a mature method for predicting the response and background of gamma-ray spectrometers [Sturner et al.(2003)]. This section describes the full end-to-end simulation chain from source generation to flight-performance assessment. The focusing optics and detector system start from independent mass models: the optics-branch mass model takes the target point source, prompt background, and delayed-background source as inputs, and records all particles transported through it that reach the focal-plane plane as the focused-source input to the detector branch; the detector branch then directly transports, in the detector/cryostat mass model of Section 1, prompt atmospheric particles, delayed decays generated from activation products according to recorded production positions, and the focused sources described above. The focused source, prompt source, and delayed source are simulated stepwise and normalized onto one time axis through Poisson-statistics-based time-axis sampling and renormalization, then the background is further suppressed by scintillator-based active veto and Compton-kinematics-based veto. The chain subsequently enters detailed background-analysis statistics, analysis of signal and background intensity variation caused by mission flight-time trajectory changes, and the overall flight-performance assessment and design-optimization recommendations. Because the focused signal and the two background streams all pass through the same response and veto definitions before any background decomposition or sensitivity statement, the subsequent results rest on a consistent event selection.

Figure 2 expands this chain by computational layer: geometry definition, prompt and activation-accumulation transport, delayed-source construction, focused-optics transport, detector-coupled signal replay, veto mechanisms, detailed background analysis, time/flight-trajectory variation, and flight-performance assessment with design-optimization recommendations. The following subsections describe each layer's inputs, operations, and outputs.

The above simulation is for a specific balloon-flight moment, longitude and latitude, and altitude. Although for an observation time such as one hour equivalent to several Monte Carlo events, this position can be regarded as constant. However, for a long-time flight on the scale of days, the altitude, longitude, and latitude obviously change significantly. This causes changes in the magnetic-rigidity cutoff for instantaneous cosmic rays and changes in the atmospheric transmission for the signal, and the activation background itself also changes with the incident cosmic-ray flux and different stages of the decay chain. This reminds us that the background-signal at a reference moment should not be linearly extrapolated in time to evaluate the time-folded flight performance over the whole working time. Therefore, we introduce a reasonable analytical flight-trajectory–time curve to evaluate the performance over the whole working cycle, and simulate multiple points on the curve to verify the rationality of the curve.

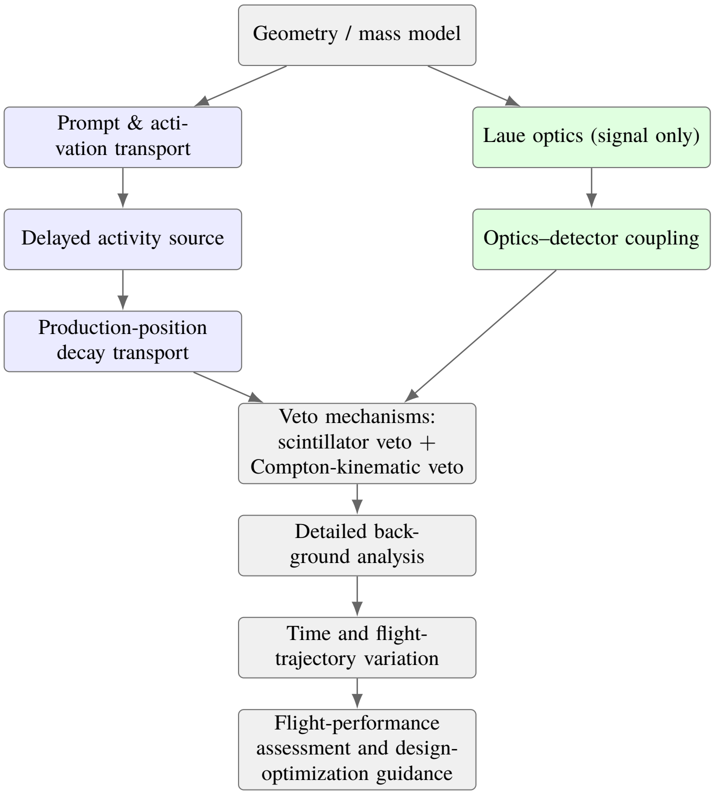

Figure 2: Simulation chain used in this work. Blue nodes are the prompt and delayed background streams, green nodes the focused-signal optics, and grey nodes the shared geometry, veto mechanisms, detailed background analysis, time/trajectory variation, and final flight-performance assessment. The focused signal and all background streams are evaluated under the same detector response and veto definitions before background decomposition, mission-time variation, and design-optimization judgments are made.

2.1 Laue optics and detector-coupled signal replay

This simulation adopts a laue-lens system based on mosaic Ge(111) crystals [Barriere et al.(2009), Barriere et al.(2011)]. Through optical design, it realizes a 10 m focal length, a multi-ring system realizes the 450keV-550keV detection band, and an effective area of 20cm**2 at 511keV; the effective area is calculated as

$$A_{\text{eff}}(E) = A_{\text{inc}} \frac{N_{\text{fp}}(E)}{N_{\text{inc}}(E)} \tag{1}$$

where A_{inc} is the Monte Carlo incident sampling area, $N_{\text{inc}}(E)$ is the number of incident photons, and $N_{\text{fp}}(E)$ is the number of photons satisfying the focusing/focal-plane interception condition. The mosaic system is adopted to make the Monte Carlo calculation convenient while satisfying the expected focusing performance; other candidate schemes are also possible for actual engineering adaptation.

In this simulation, the laue optical branch gives the focusing-performance simulation and also realizes particle transport of itself as mass.

For this purpose, the diffraction probability of photons by the crystal is no longer superposed onto photons as an efficiency parameter, but is defined as a discrete process at the Geant4 application layer [Agostinelli et al.(2003), Allison et al.(2016)]. The process takes the diffraction probability $R(|\Delta\theta|)$ according to the angular deviation of the current photon relative to the crystal plane, and converts it into a finite equivalent diffraction mean free path λ_{diff}, so that the accumulated diffraction probability when the photon crosses a crystal path ℓ reproduces this value: $1 - \exp(-\ell/\lambda_{\text{diff}}) = R(|\Delta\theta|)$, namely $\lambda_{\text{diff}} = -\ell/\ln[1 - R(|\Delta\theta|)]$; to avoid double counting with standard photoelectric absorption, R is first normalized by the unabsorbed fraction $(R/(1 - A))$ and then converted into a mean free path. After this mean free path is returned to Geant4, the diffraction process and the standard electromagnetic processes (photoelectric absorption and Compton scattering) each sample interaction lengths in the same competition framework, and the shortest one decides which process occurs first, rather than forcibly intercepting photons at the crystal boundary. Photons not selected to undergo diffraction still continue transport according to the standard electromagnetic physics, and may be absorbed, scattered, or directly transmitted.

The diffraction probability is preferentially given by interpolation in an external XOP/CRYSTAL rocking-curve map as a function of angular deviation: this curve is calculated offline once in the preprocessing stage by the CRYSTAL/diff_pat module of the XOP toolkit (called through xoppylib, with DABAX crystallographic data as input) [Sanchez del Rio & Dejus(2011)], and gives the Ge(111) reflectivity/transmissivity/absorption curves as a function of angular deviation; during transport, the angular deviation is calculated in real time for every step inside the crystal, and the current diffraction probability is obtained by interpolation on that curve, namely "offline library tabulation, probability used by angle during transport". The mosaic orientation is implemented by applying a Gaussian angular perturbation to the ideal Bragg-plane normal, with the perturbation width given by the 30 arcsec mosaic FWHM; after diffraction occurs, the outgoing direction is mirror-reflected according to the perturbed plane normal, and the photon energy is unchanged. This split in which the "process class only handles the transport interface, and the independent model class handles probability and direction decisions" follows the idea of existing crystal Bragg-reflection extensions in Geant4 [Guan et al.(2023), Reiazi et al.(2025)], and extends it to the energy band needed here.

Here the reference signal is a monochromatic 511 keV point source whose flux scale is set by the INTEGRAL/SPI upper limit on a compact 511 keV source superposed on the Galactic-centre diffuse emission ($F_0 = 10^{-4}\,\mathrm{ph\,cm^{-2}\,s^{-1}}$ above the atmosphere, Section 0.1) [Siegert et al.(2016), De Cesare(2011)], rather than by any single observed object. It is constructed as a MEGAlib far-field source [Zoglauer et al.(2006)], with the detailed construction given in Section 2.2.3. In addition to the signal we construct the prompt cosmic-ray background and the activation background as far-field sources. The transport of these three sources is simulated separately; in each run the particle flux, species, and energy crossing the focal plane are intercepted as outputs, normalized to a common equivalent observation time, and injected into the detector mass model as a parallel-beam source.

2.2 Source construction

After the focusing optics have been defined in Section 2.1, the optical input and the subsequent detector input require three source layers: the prompt atmospheric particle field incident on the payload, the delayed radioactive population accumulated in the instrument by that particle field, and the reference signal point source. These three layers are constructed separately, rather than running transport by directly invoking MEGAlib's buildup function for them at the same time, in order to reduce computational cost.

2.2.1 Prompt atmospheric source

The prompt atmospheric field is derived from EXPACS/PARMA. PARMA3.0 is an analytical model parameterized from PHITS atmospheric-shower calculations and benchmarked against terrestrial cosmic-ray measurements, and EXPACS is its public tabulation and software interface [Sato(2015), Sato(2016)]. PHITS supplies the underlying particle-transport physics: primary cosmic rays interact with air nuclei, the secondary hadrons, leptons, photons, and neutrons are transported through the atmospheric column, and the resulting fluences are parameterized by altitude, geomagnetic cutoff, solar modulation, energy, and direction. In this work EXPACS/PARMA provides the differential incident flux as a function of particle species, kinetic energy, zenith angle, geographic position, altitude, cutoff rigidity, and solar activity, and is used only to construct the external radiation fields; detector response, vetoing, topology selection, and mission-time counting are applied downstream. The reference source definitions use latitude 34°, longitude 100°, altitude 38 km, vertical cutoff rigidity $R_c = 11.6\,\mathrm{GV}$, and the 2025-08-31 solar condition corresponding to $W = 118.3$.

The prompt source includes photons, neutrons, electrons, positrons, protons, alpha particles, and negative and positive muons. For each particle family the EXPACS/PARMA differential spectrum is converted into a MEGAlib far-field area source covering the full zenith range $\theta = 0°$–$180°$ and full azimuth range $\phi = 0°$–$360°$. The zenith dependence is represented by 20 differential equal-μ angular bins, where $\mu = \cos\theta$ and each bin subtends $\Delta\Omega = 0.6283185307\,\mathrm{sr}$: the first ten bins cover down-going particles, the last ten up-going or albedo-

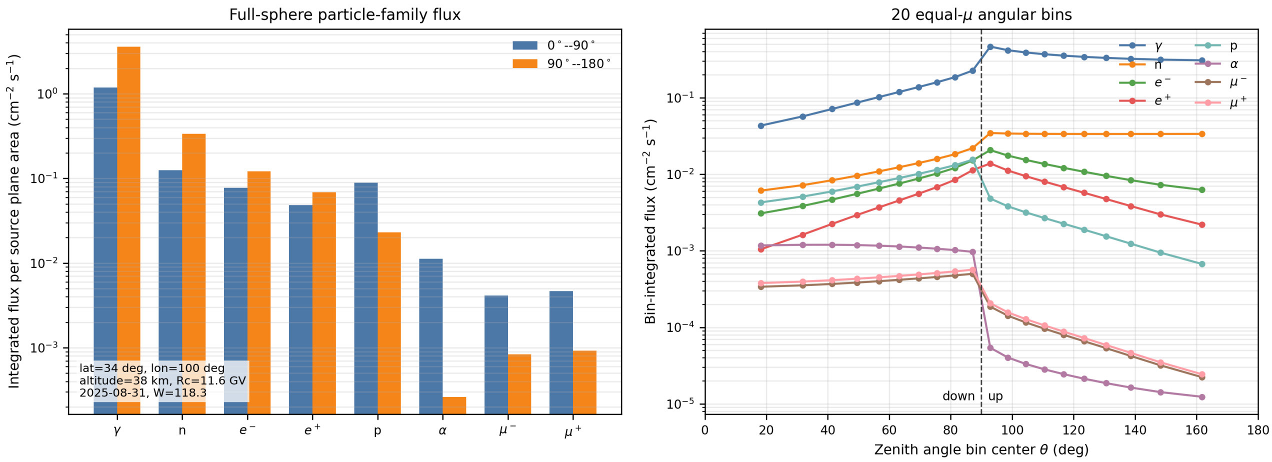

Figure 3: EXPACS/PARMA full-sphere atmospheric flux field used to generate the prompt and activation-production transport inputs. Left: energy- and solid-angle-integrated fluxes for the eight transported particle families, separated into down-going and up-going hemispheres. Right: bin-integrated flux in the 20 equal-μ differential zenith bins used by the MEGAlib far-field source. The plot is generated from the source-definition inputs for latitude $34°$, longitude $100°$, altitude $38\,\text{km}$, $R_c = 11.6\,\text{GV}$, and $W = 118.3$.

like particles. Both hemispheres are retained in the prompt transport and in the activation-production transport, so upward albedo particles are not discarded at source construction. Figure 3 shows the resulting full-sphere flux field.

The integrated source-plane fluxes are approximately 4.80, 0.462, 0.199, 0.117, 0.112, 0.0115, 0.0050, and $0.0056\,\text{cm}^{-2}\,\text{s}^{-1}$ for γ, n, e^-, e^+, p, α, μ^-, and μ^+, respectively; the non-photon families together carry about $0.91\,\text{cm}^{-2}\,\text{s}^{-1}$, or 16% of the total. If Monte Carlo counts were allocated strictly in proportion to flux, the detector and activation statistics of these minority families would be poor, so the non-photon families are generated with replicated samples and de-weighted in the rate normalization. All particles are then transported through the detector/cryostat mass model with Geant4/MEGAlib [Agostinelli et al.(2003), Allison et al.(2016), Zoglauer et al.(2006)]; the prompt run and the activation-production run each generated 25,210,216 incident primaries. Each particle species is normalized with its own transported exposure,

$$r_{\text{event},j} = \left(\sum_i \mathcal{T}_{ij} \right)^{-1}, \tag{2}$$

where $\mathcal{T}_{ij}$ is the transported exposure of subsample i for species j. This per-species normalization is required because the photon and non-photon streams have different sampling and exposure structures; the normalization check enforces $r_{\text{event},j} \sum_i \mathcal{T}_{ij} = 1$ for every species, so over-sampling changes the Monte Carlo variance but not the expected physical rate.

2.2.2 Delayed activation source

For the delayed simulation, a separate activation-production stream is run by enabling BUILDUP during the prompt-background transport. This activation-production stream is not counted as an immediate detector background. Instead, it records the isotope, material volume, and production position of each radioactive product. These records are converted through the production–decay accumulation equation into a radioactive population fixed to a chosen reference time during the flight, with ground-state half-lives checked against NUBASE2020 [Kondev et al.(2021)]. For isotope k,

$$A_k(t_{\text{flight}}) = P_k \left[1 - \exp\left(-\lambda_k t_{\text{flight}} \right) \right], \qquad \lambda_k = \frac{\ln 2}{t_{1/2,k}}, \tag{3}$$

where P_k is the production rate inferred from the activation-production transport and $t_{\text{flight}} = 15\,\text{d}$ for the reference state. The fixed delayed total activity used by the reference delayed-source construction is $85.45\,\text{Bq}$.

The position information does not come from the aggregate isotope store alone. The standard store records volume names and isotope counts; the production-position construction adds a custom recording hook in the MEGAlib stepping-action layer. Whenever a delayed isotope is stored during activation buildup, the hook writes a production-position record at the same moment, including the logical volume name, pre-step position (x, y, z), isotope identifier $1000Z + A$, excitation energy, time, process, and track-ancestry information. The fixed day-15 population is then matched to these records by volume, isotope, and excitation state, so the delayed source can use the simulated production coordinates rather than only volume-integrated activities. Figure **??** in Appendix **??** shows the resulting position distribution: the activity-weighted map is dominated by CsI-shield activation and ^{128}I, with smaller contributions from Cu, Al, W, Mg, and Cs isotopes. It is a source-construction diagnostic, not a detector-count map; detector response is applied only after the delayed decays are transported through the common geometry and event selection.

Whether the production positions are worth preserving can be tested from the same table. Delayed decays are emitted locally, so downstream attenuation, active-shield veto probability, and event topology all depend on where the isotope was produced, and different incident families—with their different energy-loss, stopping, and capture histories—need not deposit their activation in the same materials. To quantify this, the weighted production rows are grouped by incident particle family and material category, and the activity-normalized category distributions of two families a and b are compared with the total-variation distance

$$D_{\mathrm{TV}}(a, b) = \frac{1}{2} \sum_c |f_{a,c} - f_{b,c}| \,, \tag{4}$$

where $f_{a,c}$ is the fraction of delayed activity from family a produced in material category c. In the fixed day-15 table, neutron-induced products carry about 82.5 Bq, μ^--induced products 2.75 Bq, proton-induced products 0.15 Bq, and α-induced products 0.022 Bq, with smaller μ^+ and e^+ terms. The neutron-induced activity remains dominated by the CsI shield, whereas the μ^- component is qualitatively different: Cu/support material, Al shells, and W-bearing passive material account for 32%, 29%, and 21% of the μ^--induced activity, respectively (Figure **??** in Appendix **??**). A volume-integrated or axisymmetric radial source could preserve the total activity, but it would erase these incident-family, material, nuclide, and coordinate correlations; the production-position construction keeps them in the source before detector transport.

The delayed-decay source is therefore emitted from sampled production positions rather than from an axisymmetric distribution. Let row j in the weighted production table have activity weight w_j, with total activity $A = \sum_j w_j$. During construction, M production positions are drawn with replacement using probability $p_j = w_j/A$, and each sampled point source is assigned the same flux A/M. The expected activity assigned to row j is

$$\mathbb{E}\left[N_j \frac{A}{M}\right] = M p_j \frac{A}{M} = w_j, \tag{5}$$

so the estimator preserves the total activity and is unbiased for the weighted spatial distribution. Here M is the number of equal-activity sampled point sources; it is a sampling size, not the number of physical production rows. The reference delayed source preserves the fixed total activity at the source-definition level and is replayed through the same detector/cryostat geometry as the prompt stream. The nominal calculation transports 10^6 delayed decays; the corresponding Cosima live time is only the transport record used to generate the decay sequence and is not used as the physical balloon exposure time.

Table 1 summarizes the source layers used in the current detector-coupled calculation. It is placed in the source-model section because this section defines incident particle fields and delayed radioactive populations; detector vetoes and topology selections are applied only in the subsequent selection section.

This production-position representation does not create one sampled point source for every production row. It is a statistical sampling approximation whose adequacy is evaluated separately at the selected-rate level in Section **??**. Source-list checks verify activity conservation and source serialization; selected-rate behavior is tested separately after detector transport and event selection.

Table 1: Background-source layers used by the current detector-coupled chain. The table defines source construction only; anticoincidence, topology selection, and the line window are applied later in Section 2.3.

Layer	Physical role	Current implementation
Prompt atmospheric transport	Direct detector hits from atmospheric secondaries	EXPACS/PARMA full-sphere source definitions at 34° latitude, 100° longitude, 38 km altitude, $R_c = 11.6\,\mathrm{GV}$, and $W = 118.3$; γ, n, $e^\pm$, p, α, and $\mu^\pm$ over 20 equal-μ differential zenith bins including both down-going and up-going hemispheres; 25,210,216 generated primaries.
Activation production	Radioisotope production and production-position history	Same atmospheric field transported as a BUILDUP activation-production stream, again retaining both hemispheres and all eight particle families; 25,210,216 generated primaries.
Fixed delayed population	Radioactive activity at the adopted day-15 state after continuous exposure	Production–decay accumulation with NUBASE-checked ground-state half-lives; total fixed activity $85.45\,\mathrm{Bq}$.
Production-position decay transport	Delayed decay particles emitted from sampled production positions	Equal-activity sampled point sources from the weighted production table; reference transport uses 10^6 delayed decays per source sampling.

2.2.3 Reference point-source construction

The focused signal stream is defined by a monochromatic 511 keV reference point source. Its flux scale is set not by any single observed object but by the INTEGRAL/SPI upper limit on a compact 511 keV source superposed on the Galactic-centre diffuse emission, $F_0 = 10^{-4}\,\mathrm{ph\,cm^{-2}\,s^{-1}}$ above the atmosphere (Section 0.1) [Siegert et al.(2016), De Cesare(2011)].

Because such a source lies at astronomical distance, it is modelled as an ideal on-axis point: in the optics run it is a monochromatic 511 keV far-field beam whose photons arrive parallel to the optical axis and are sampled uniformly over the incident aperture A_{inc} of Eq. (1). No finite angular size is assigned to the source itself; the angular structure that reaches the focal plane is produced entirely by the lens—the 30 arcsec mosaic spread, the XOP/CRYSTAL rocking curve, and the $f = 10\,\mathrm{m}$ focusing geometry—so that the replayed beam converges to within about 0.4° of the axis and forms a focused spot rather than a single ray. Three independent seeds of 5×10^4 primaries each give 1.5×10^5 transported photons, of which 37,256 are diffracted and cross the focal plane; 37,194 of these (99.8%) fall within the $r = 1.898\,\mathrm{cm}$ Be entrance window, with focused-spot radii $r_{90} = 1.03\,\mathrm{cm}$ and $r_{99} = 1.25\,\mathrm{cm}$. The within-window crossings form the event list that is replayed through the Be aperture into the detector/cryostat model with the recorded positions and directions.

The normalization ties this replay to the reference flux. At the detector entrance the focused stream carries the rate $F_0\,A_{\mathrm{eff}}(511\,\mathrm{keV})\,T$, where $A_{\mathrm{eff}}(511\,\mathrm{keV}) = 20.1\,\mathrm{cm^2}$ is the on-axis effective area and $T = 0.739$ is the reference 511 keV atmospheric transmission at the day-15 trajectory anchor; numerically this is $1.48 \times 10^{-3}\,\mathrm{s^{-1}}$ at the reference flux. Equivalently, the generating run corresponds to an observation time $N_{\mathrm{fp}}/[F_0\,A_{\mathrm{eff}}\,T]$, and each replayed event carries the inverse of that exposure as its rate weight; in the intermediate cut-flow bookkeeping of Section 2.3 the same sample appears as a unit replay before this scale is applied. After the common detector response and event selection, the surviving signal rate is $1.19 \times 10^{-3}\,\mathrm{s^{-1}}$, corresponding to a selected effective area of $S_{W_{511}}/F_0 = 11.9\,\mathrm{cm^2}$. Because the optics run contains only source photons, the replayed stream is signal-pure; all backgrounds enter through the streams of Section 2.2.

The effective area quoted above is the on-axis value, representing best-case pointing. A point source offset from the optical axis, or residual attitude jitter during flight, reduces and shifts the focused response through the finite angular acceptance (field of view) of the Laue lens [Barriere et al.(2009), Weidenspointner et al.(2006)]; this off-axis dependence is treated as a pointing systematic on the signal effective area rather than as a change to the on-axis reference construction.

2.3 Detector event selection and veto definitions

The analysis window is

$$W_{511} = [510.79, 511.21]\,\mathrm{keV}. \tag{6}$$

The first selection layer is the active-shield veto. Candidate events are grouped on a common Poisson time axis and are rejected if the summed CsI shield energy in the coincidence window exceeds the analysis threshold. For the baseline CsI configuration the veto threshold is $50\,\mathrm{keV}$ and the coincidence window is $1\,\mu$s. The shield sum is evaluated from the CsI active-shield volumes in post-processing. Native detector trigger/veto blocks are present in the geometry description, but the quantitative rate definition is this common post-processing definition, so prompt, delayed, and focused-signal streams use the same coincidence window and threshold.

The second layer is a topology/field-of-view consistency test on the TES hits that survive the active-shield gate. Baseline event classes are kept explicit: single-site events, valid reconstructed multi-site events, and unreconstructed multi-site events. Single-pixel TES events are retained as calorimetric line candidates. For a two-hit event, both possible first-scatter orders are tested under the assumptions of a single Compton scatter followed by full absorption in the measured TES hits. For an assumed order with first deposited energy E_1 and remaining scattered-photon energy E_{sc}, the kinematic angle is computed from

$$\cos\theta_C = 1 - m_e c^2 \left(\frac{1}{E_{\mathrm{sc}}} - \frac{1}{E_1 + E_{\mathrm{sc}}} \right). \tag{7}$$

The event is kept if either order gives a physically valid first-scatter cone that intersects the side Be-window disk. For higher multiplicities the current implementation enumerates candidate hit orders up to six TES hits, selects the best sequence using the unweighted Compton sequence residual

$$Q = \sum_{i=1}^{n-2} \left(\cos\theta_{C,i} - \hat{\mathbf{u}}_{i-1,i} \cdot \hat{\mathbf{u}}_{i,i+1} \right)^2, \tag{8}$$

and then applies the same first-cone intersection test to the first reconstructed scatter. If no valid sequence is found, the event is classified as unreconstructed. Such events are retained in the baseline selected-rate definition but are not called field-of-view passing. The line-window result therefore does not gain sensitivity from rejecting ambiguous reconstruction failures. This policy is not a conservative lower bound on the net sensitivity: in the current W_{511} sample, a single-site-only diagnostic alternative that vetoes all multi-site candidates would retain 59.4% of the focused signal and 68.2% of the selected background, reducing the simple $S/\sqrt{B}$ figure to 0.720 of the baseline value. The baseline selection therefore treats ambiguous multi-hit events as a reconstruction-model boundary that must be validated, not as a sensitivity gain from aggressive event rejection.

To connect this selection chain explicitly to the finite Monte Carlo sample, Table 2 gives the current event-level W_{511} cut-flow obtained by lightweight replay of the simulated event catalogues. The table is not a new Geant4/MEGAlib transport. It replays the current event catalogues with the same TES energy window, CsI veto, and Compton-kinematic/field-of-view consistency checks; uncertainties are Monte Carlo quadrature sums over event weights. The current event-selection records first fix the energy window and then report raw, active-shield, and topology stages, so the raw stage should be read as the pre-veto sample inside the line window, not as an all-energy incident sample.

The detector-level time-axis construction used to validate this veto bookkeeping is a Poisson superposition of independent streams. For stream k with event-catalogue rates r_{kj} and total rate $R_k = \sum_j r_{kj}$, a validation 81 points from 0 to 20 d with 0.25 d spacing; endpoint points carry half-widths in the count integration. The

Table 2: event-level W_{511} line-window cut-flow.

Stream	Stage	Events	Rate / s^{-1}	Statistical uncertainty / s^{-1}
Prompt background	Line-window pre-veto	161	1.1877×10^{-1}	1.15×10^{-2}
Prompt background	After CsI veto	60	4.0712×10^{-2}	5.26×10^{-3}
Prompt background	Topology/FOV selected	54	3.6641×10^{-2}	4.99×10^{-3}
Delayed activation	Line-window pre-veto	54	4.6354×10^{-3}	6.31×10^{-4}
Delayed activation	After CsI veto	33	2.8327×10^{-3}	4.93×10^{-4}
Delayed activation	Topology/FOV selected	30	2.5752×10^{-3}	4.70×10^{-4}
Focused signal	Line-window pre-veto	30323	8.1527×10^{-1}	4.68×10^{-3}
Focused signal	After CsI veto	30323	8.1527×10^{-1}	4.68×10^{-3}
Focused signal	Topology/FOV selected	29715	7.9892×10^{-1}	4.63×10^{-3}

The focused-signal rows are unit-replay rates used to show veto survival behavior; the point-source signal rate in Table 3 is scaled separately by the Laue effective area, reference flux, and atmospheric transmission.

trajectory is a synthetic reference profile, not balloon telemetry. For time-grid point t_i in days,

$$h_i = h_0 + \Delta h \sin\left[\frac{2\pi(t_i - 15)}{5}\right], \tag{9}$$

$$\varphi_i = \varphi_0 + \Delta\varphi \sin\left[\frac{2\pi(t_i - 15)}{7}\right], \tag{10}$$

$$\ell_i = \ell_0 + \Delta\ell \sin\left[\frac{2\pi(t_i - 15)}{9}\right], \tag{11}$$

where $h_0 = 38.75\,\text{km}$, $\Delta h = 2.5\,\text{km}$, $\varphi_0 = 34°$, $\Delta\varphi = 0.25°$, $\ell_0 = 100°$, and $\Delta\ell = 0.25°$. The resulting ranges are 36.25–41.25 km in altitude, 33.75°–34.25° in latitude, and 99.75°–100.25° in longitude.

The atmospheric depth and the 511 keV transmission scale are

$$X(h_i) = X_{38} \exp\left[-\frac{h_i - 38\,\text{km}}{H}\right], \qquad X_{38} = 3.87\,\text{g cm}^{-2}, \quad H = 6.8\,\text{km}, \tag{12}$$

$$T_i = \exp[-\mu_{\text{eff}} X(h_i)], \qquad \mu_{\text{eff}} = -\frac{\ln T_{15}}{X(h_0)}, \tag{13}$$

with $T_{15} = 0.739$. Thus the focused-science scale is $f_{\text{atm},i} = T_i/T_{15}$. The prompt atmospheric source scale is

$$R_{c,i} = 11.0 - 0.08(\varphi_i - \varphi_0) + 0.03(\ell_i - \ell_0), \tag{14}$$

$$f_{\text{p},i} = \exp\left[\frac{X(h_i) - X(h_0)}{30\,\text{g cm}^{-2}}\right]\left(\frac{11.0}{R_{c,i}}\right)^{0.2}. \tag{15}$$

The trajectory rigidity model is centered on a rounded value, and $f_{\text{p},i}$ enters only through the day-15–anchored ratio $(11.0/R_{c,i})^{0.2}$, so this central value affects the mission-time variation but not the absolute prompt rate, which is fixed by the $R_c = 11.6\,\text{GV}$ source definition. The same scalar production driver is used for delayed activation before isotope-by-isotope decay buildup is solved. For a delayed-production element a with decay constant λ_a and fixed day-15 activity $A_{a,15}$,

$$P_{a,15} = \frac{A_{a,15}}{1 - \exp(-\lambda_a t_{15})}, \tag{16}$$

$$\frac{dN_a}{dt} = P_{a,15} f_{\text{p}}(t) - \lambda_a N_a, \qquad A_a(t) = \lambda_a N_a(t). \tag{17}$$

The numerical fold anchors the solution back to $A_{a,15}$ at day 15 and sums the activity over the nuclide/material/position production elements.

This delayed-activity treatment assumes that the normalized activation-product distribution is unchanged over the small trajectory range, and that the trajectory changes only the scalar production driver. The verification described here checks the internal implementation of that model but does not prove physical invariance of the production distribution. If dedicated activation transports at the trajectory center and extrema show material or nuclide redistribution, the scalar driver should be replaced by interpolation over production vectors,

$$P_a(t_i) = \sum_g w_g(h_i, \varphi_i, \ell_i, R_{c,i}) P_a^{(g)}, \qquad \sum_g w_g = 1, \tag{18}$$

followed by the same decay ODE.

Implementation checks recomputed the trajectory formula at the center point and at the two altitude extrema. The reference atmospheric transmission is $T = 0.739$ at the day-15 anchor, and the high- and low-altitude probes give $T = 0.811$ and 0.646, respectively. Reconstructing the W_{511} prompt, delayed, background, and signal rates from the day-15 rates and these scales reproduces the mission-fold summary. Reweighting the particle families with local PARMA spectra at the same trajectory probes shifts the normalized nuclide–volume activity distribution by less than 6×10^{-3} in total-variation distance. This supports using a scalar delayed production/activity fold for the present small trajectory range at the particle-family reweighting layer. It is not a full proof of distribution invariance: same-particle energy/angle-dependent activation cross sections and position-transport shifts would require dedicated low-, reference-, and high-altitude activation-production transports.

In the mission-time fold, the selected day-15 rates are folded over the trajectory grid as

$$B_i = L_i \left(B_{\mathrm{p},15} f_{\mathrm{p},i} + B_{\mathrm{d},15} f_{\mathrm{d},i} \right), \tag{19}$$

$$S_i = L_i S_{15} f_{\mathrm{atm},i}, \tag{20}$$

$$L_i = \exp[-(R_{\mathrm{p},i} + R_{\mathrm{d},i})\tau], \tag{21}$$

where $f_{\mathrm{p},i}$ is the prompt atmospheric scale, $f_{\mathrm{d},i}$ is the delayed-activity scale, $f_{\mathrm{atm},i}$ is the 511 keV atmospheric transmission scale, and L_i is the accidental coincidence live factor. Here $R_{\mathrm{p},i}$ and $R_{\mathrm{d},i}$ are the prompt and delayed event rates entering the coincidence grouping at that mission point. The factor L_i assumes a one-sided τ gate; a symmetric $\pm\tau$ condition would instead give $\exp[-2(R_{\mathrm{p},i} + R_{\mathrm{d},i})\tau]$, but for the present rates both forms remain very close to unity, so the folded counts are insensitive to this choice.

The time-dependent significance is computed as a cumulative counting statistic. For a reference flux F_0, the cumulative significance is

$$Z(t_m) = \frac{N_s(t_m)}{\sqrt{N_b(t_m)}}, \tag{22}$$

where

$$N_s(t_m) = \sum_{i \le m} S_i \Delta t_i, \qquad N_b(t_m) = \sum_{i \le m} B_i \Delta t_i \tag{23}$$

are the time-integrated selected signal and background counts after mission-time scaling and live-factor corrections. For comparison among analysis variants, the 3σ flux threshold is scaled as

$$F_{3\sigma}(t_m) = F_0 \frac{3}{Z(t_m)}. \tag{24}$$

This Gaussian counting convention is used consistently for the line-window results and secondary comparisons, but it is a diagnostic selected-rate metric, not a final flight discovery likelihood. For the present 20 d reference case, the signal-to-background ratio is about 3%; a 1% Gaussian uncertainty on the background normalization alone would reduce the approximate significance from about 7.8 to about 2.8. A flight analysis would therefore require off-source or control-band background constraints and nuisance-parameter profiling. For the primary 20 d line-window result, the folded counts are about 6.8×10^4 background counts and 2.0×10^3 signal counts at F_0, so this approximation is used only in the high-count selected-rate regime. The isolated upstream Ge-proxy result with zero selected events is reported separately with a Poisson zero-count upper limit. The detector-selection summary also records a day-15 direct expectation. The primary result does not use that constant-rate direct value; it uses the mission-time fold and the final cumulative significance. The distinction is small for the present line-window case, but keeping it explicit is necessary for comparing analysis configurations.

3 Results

The recovered log fragments around the original detailed results tables are structurally mixed. This compile-ready copy therefore keeps the confirmed primary selected-rate numbers in a compact recovery table and leaves the original detailed table reconstruction to manual editing.

Table 3: Recovered primary unresolved-line selected-rate result.

Quantity	Value	Note
Day-15 selected background	$3.92 \times 10^{-2}\,\mathrm{s}^{-1}$	Prompt + delayed
Day-15 selected signal at F_0	$1.19 \times 10^{-3}\,\mathrm{s}^{-1}$	$F_0 = 10^{-4}\,\mathrm{ph\,cm^{-2}\,s^{-1}}$
Prompt background fraction	93.4%	Dominant component
20 d diagnostic Z	7.80	Counting metric
$F_{3\sigma}(20\,\mathrm{d})$	$3.85 \times 10^{-5}\,\mathrm{ph\,cm^{-2}\,s^{-1}}$	Reference model only

Recovery gap note

The original recovered log fragments become structurally inconsistent after this point. The broad-window diagnostic paragraph and several nearby transition lines are omitted from this compile-ready reading copy; see the raw partial source and recovery manifest.

4 Discussion and conclusions

The main result is that the reference production-position delayed-source calculation gives a 20 d diagnostic 3σ point-source threshold of $3.85 \times 10^{-5}\,\mathrm{ph\,cm^{-2}\,s^{-1}}$ for an unresolved line in the modeled focused TES line spectrometer. This threshold is obtained without using the focused-spot spatial analysis as the primary result. It follows from three combined effects: optical concentration, a sub-keV energy window, and a detector-coupled veto/topology selection that suppresses prompt and delayed backgrounds in the same selection space as the signal.

This calculation should be compared with existing 511 keV instruments as a pointed, narrow-line, compact-source estimate, not as an all-sky imaging forecast. INTEGRAL/SPI and COSI provide the reference measurements for mapping the diffuse 511 keV sky and the broader Galactic diffuse emission [Siegert et al.(2016), Kierans et al.(2020), Siegert et al.(2020), Yoneda et al.(2025), Karwin et al.(2023)]. Peer-reviewed COSI calibration and simulation studies provide the more relevant technical comparison for detector response, event reconstruction, and balloon-background validation [Beechert et al.(2022), Sleator et al.(2019), Gallego et al.(2025a), Gallego et al.(2025b), Ciabattoni et al.(2025)]. The TES Laue-lens concept considered here is complementary: it trades sky coverage for a concentrated beam and a sub-keV line window. The relevant comparison class is compact 511 keV targets, unresolved-central-excess searches, and transient follow-up observations with known or constrained source positions. The same focused aperture is not a substitute for measuring the full diffuse bulge and disk luminosity, because it samples only a small solid angle.

The production-position delayed-source treatment addresses a specific approximation in radial-profile compression. Although delayed activation contributes only about 6.6% of the selected line-window background in the present calculation, it is spatially and materially structured. Preserving sampled production positions prevents artificial azimuthal or radial redistribution from becoming an unquantified systematic. The delayed component has passed a minimum selected-rate convergence check using four independent production-position samplings, but this should not be read as a complete delayed-background systematic envelope.

The background composition indicates the first engineering parameters to test, rather than a completed optimization. The selected line-window rate remains dominated by prompt atmospheric streams, so shield geometry, side-entry collimation, material near the focal plane, topology selection, and multi-pixel reconstruction are the most direct handles on the present background. Delayed activation is smaller but remains spatially and materially structured, so source-position preservation is retained in the rate calculation. The dominance of iodine activation in the CsI shield also motivates a future shield-material trade study.

4.1 Limitations and future validation

The selected-rate result has five immediate boundaries. First, the detector/cryostat model is an engineering proxy, not a final flight CAD model. Second, quantitative active-shield rejection is applied in the common detector-level post-processing selection rather than by production-time trigger/veto filtering. Third, the mission-time calculation is a rate-level fold, not a new particle transport at each trajectory time bin. Fourth, the custom Compton/field-of-view veto is a detector-level topology and kinematic consistency check, not yet a full Revan/Mimrec reconstruction, and the retained reconstruction-failure class is a baseline acceptance choice rather than a conservative sensitivity bound. Fifth, the event-level analysis has made the cut-flow, broad-window spectra, veto behavior, and activation selected-event decomposition traceable, but selected delayed state/metastable labels, the merged four-sampling per-event decomposition, post-veto energy-window staging, multi-pixel residual distributions, and veto-threshold stability scans are still not closed by the current outputs.

More generally, the tabulated 3σ flux thresholds are statistical selected-rate thresholds for the reference source, mass, and response model. They do not include a systematic-uncertainty envelope for atmospheric-flux normalization, activation cross sections or physics-list choices, material and geometry tolerances, detector energy-response uncertainty, veto-threshold calibration, reconstruction-efficiency systematics, or background-estimation nuisance parameters. A publication-level flight sensitivity should replace the diagnostic $S/\sqrt{B}$ metric with a spatial-spectral likelihood, following standard profile-likelihood practice [Cowan et al.(2011)].

The upstream Laue-lens calculation in Section **??** should be read as a boundary calculation for an equal-volume/equal-mass active-Ge proxy, not as a finalized optics-background budget. It completes full-particle prompt and activation-production transport for the proxy, and the isolated Ge-proxy delayed component gives zero selected W_{511} events in the present transport. The prompt self-background contribution is still not part of the primary detector-background budget because it needs source separation or subtraction from the ordinary detector/cryostat prompt background, and explicit lens support hardware is not yet modeled.

Spatial analyses such as r_{90} and multi-annulus likelihood estimates are diagnostic cross-checks, but they are not yet a finalized nuisance-profile likelihood. The quoted sensitivity therefore remains the line-window selected-rate result, not a spatially optimized likelihood result.

Before this rate study can support a flight-performance statement, four validation blocks remain: a Revan/Mimrec or equivalent reconstruction cross-check of the retained multi-hit subset, including residual distributions and threshold stability; optics self-background transport with explicit lens tiles and support hardware at final statistics; a final cryostat/DR geometry and detector-response update, including TES saturation and pile-up constraints; and a spatial-spectral likelihood that profiles background nuisance parameters. The event-level tables added here should be read as manuscript support evidence and audit entry points before those validations, not as a complete systematic-uncertainty envelope.

4.2 Conclusions

We have presented a detector-coupled background and sensitivity study for a balloon-borne pointed 511 keV TES Laue-lens telescope. The reference calculation couples a Ge(111) Laue optical response to a TES focal-plane mass model, prompt atmospheric transport, delayed activation, active-shield vetoing, topology/field-of-view selection, mission-time rate folding, and counting-significance evaluation. In the primary 510.79–511.21 keV unresolved-line window, the day-15 selected rates are $B \simeq 3.92 \times 10^{-2}\,\mathrm{s}^{-1}$ and $S \simeq 1.19 \times 10^{-3}\,\mathrm{s}^{-1}$ at $10^{-4}\,\mathrm{ph\,cm^{-2}\,s^{-1}}$; the mission-time fold gives $Z_{20\mathrm{d}} \simeq 7.80$ and $F_{3\sigma}(20\mathrm{d}) \simeq 3.85 \times 10^{-5}\,\mathrm{ph\,cm^{-2}\,s^{-1}}$. The result provides a detector-coupled selected-rate basis for continued evaluation of focused TES spectroscopy as a complementary 511 keV technique for compact-source and unresolved-central-excess searches, while identifying prompt atmospheric background suppression, delayed-activation systematic validation, and reconstruction validation as the main next validation tasks.

Data and software availability

Before publication, a versioned public repository or institutional archive should provide the geometry and source definitions, random seeds, analysis and figure-generation scripts, LaTeX source, and derived validation products needed to reproduce the tables and figures in this paper. This compile-ready recovery copy is not the original final manuscript.

Declarations

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